Mathematics practice booklet: Grade 5 expectations

Grade 5 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 5 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: 5.OA.A.1, 5.OA.A.2, 5.OA.B.3

A worked example

Work out 4 x (6 + 3) - 5. Then write the expression for: add 6 and 3, then multiply the sum by 4.

1. Brackets first: 6 + 3 = 9.
2. Then multiply: 4 x 9 = 36.
3. Then subtract: 36 - 5 = 31.
4. Expression for the words: 4 x (6 + 3).

31; 4 x (6 + 3)

Questions

1. Evaluate: 18 - (4 + 5)

2. Evaluate: 3 x (12 - 7) + 8

3. Evaluate: {20 - [3 x (2 + 1)]} / 2

4. Write as an expression: three times the sum of 8 and 4.

5. Without calculating, is 2 x (45 + 17) bigger or smaller than 45 + 17?

6. Rule A: start at 0 and add 3. Rule B: start at 0 and add 6. Write the first five numbers of each.

7. How does each number in Rule B compare with the matching number in Rule A?

8. Write the ordered pairs from the last two questions (A first, B second).

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9. A taxi charges AED 10 to start and AED 2 for each km. Write a rule for the cost and find the cost of 6 km.

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10. Insert brackets to make this true: 2 + 3 x 4 = 20

Section 2: Shapes, measuring and space

Standards practised: 5.MD.A.1, 5.MD.C.3, 5.MD.C.4, 5.MD.C.5, 5.G.A.1, 5.G.A.2, 5.G.B.3, 5.G.B.4

A worked example

A fish tank is 50 cm long, 30 cm wide and 40 cm high. Find its volume in cubic centimetres and in litres (1 litre = 1,000 cubic cm).

1. Volume = length x width x height.
2. 50 x 30 = 1,500, then 1,500 x 40 = 60,000 cubic cm.
3. 60,000 / 1,000 = 60 litres.

60,000 cubic cm, which is 60 litres

Questions

1. A box is 5 cm by 4 cm by 3 cm. What is its volume?

2. A cube has edges of 4 cm. What is its volume?

3. A box has a volume of 120 cubic cm. Its base is 10 cm by 4 cm. How tall is it?

4. Convert 2.5 m to centimetres.

5. Convert 3,200 mL to litres.

6. Point A is at (2, 3) on a grid. Starting at the origin, describe how to get there.

7. Plot the points (1, 1), (4, 1), (4, 3) and (1, 3) and join them in order. What shape do they make, and what is its area?

8. Is every square also a rectangle? Explain.

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9. Is every rectangle also a square? Explain.

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10. A pool is 10 m by 5 m with water 2 m deep. How many cubic metres of water fill it?

Section 3: Data, graphs and chance

Standards practised: 5.MD.B.2, 5.NF.A.1, 5.NF.B.3

A worked example

Water in 5 cups measures 1/2, 1/4, 1/4, 1/2 and 1/2 of a litre. If all the water is poured evenly into the 5 cups, how much does each hold?

1. Add: 1/2 + 1/4 + 1/4 + 1/2 + 1/2 = 2/4 + 1/4 + 1/4 + 2/4 + 2/4 = 8/4 = 2 litres.
2. Share equally between 5 cups: 2 / 5 = 2/5 of a litre.

2/5 of a litre

Questions

1. A line plot shows 4 dots at 1/4 and 2 dots at 1/2. What is the total?

2. Four friends have 3, 5, 6 and 10 stickers. They put them together and share them equally. How many stickers does each get?

3. A line plot shows 4 cups of water: 2 dots at 1/2 litre, 1 dot at 1/4 litre and 1 dot at 3/4 litre. How many litres of water are there in all?

4. If the water in the last question is poured evenly into the 4 cups, how much is in each cup?

5. Is the answer to the worked example, 2/5 of a litre, more or less than 1/2 litre? Explain.

6. A line plot of eggs per nest: 2 nests have 3 eggs each, 5 nests have 4 eggs each and 3 nests have 5 eggs each. How many nests are there?

7. Using the same data, how many eggs are there in all?

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8. In that data, how many eggs does a nest most often have?

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9. In the worked example, how much more water is in all the 1/2 litre cups than in all the 1/4 litre cups?

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10. Check the worked example: if each of the 5 cups holds 2/5 litre, how much water is that altogether? Does it match the total?

Section 4: Numbers and calculating

Standards practised: 5.NBT.A.1, 5.NBT.A.2, 5.NBT.A.3, 5.NBT.A.4, 5.NBT.B.5, 5.NBT.B.6, 5.NBT.B.7, 5.NF.A.1, 5.NF.B.3, 5.NF.B.4, 5.NF.B.5, 5.NF.B.7

A worked example

What is 2/3 + 1/4? What is 2/3 x 3/5?

1. Unlike denominators: find a common one, 12. 2/3 = 8/12 and 1/4 = 3/12.
2. 8/12 + 3/12 = 11/12.
3. Multiplying: top times top and bottom times bottom: 2 x 3 = 6 and 3 x 5 = 15, giving 6/15 = 2/5.

11/12 and 2/5

Questions

1. What is 10 times 3.4? What is one tenth of 3.4?

2. Write 4.075 in words.

3. Compare using <, > or =: 0.62 and 0.6

4. Round 7.846 to the nearest hundredth.

5. Multiply: 346 x 27

6. Divide: 1,248 / 24

7. Add: 12.45 + 3.8

8. Subtract: 20 - 7.35

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9. Calculate: 3/4 - 1/6

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10. 4 friends share 3 pizzas equally. How much pizza does each get? Write it as a fraction.

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11. 3 friends share 1/4 of a pizza equally. What fraction of the whole pizza does each friend get?

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12. Without multiplying, is 5 x 2/3 greater or less than 5? Explain.

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. 9
2. 23
3. 5.5 (inside: 2 + 1 = 3, 3 x 3 = 9, 20 - 9 = 11, 11 / 2)
4. 3 x (8 + 4)
5. Bigger: it is twice the sum.
6. A: 3, 6, 9, 12, 15. B: 6, 12, 18, 24, 30.
7. It is twice as big.
8. (3, 6), (6, 12), (9, 18), (12, 24), (15, 30)
9. Cost = 10 + 2 x km; 6 km costs AED 22.
10. (2 + 3) x 4 = 20

Section 2: Shapes, measuring and space

1. 60 cubic cm
2. 64 cubic cm
3. 3 cm
4. 250 cm
5. 3.2 litres
6. Go 2 units right, then 3 units up.
7. A rectangle, 3 units by 2 units: area 6 square units
8. Yes: a square has four right angles and opposite sides equal.
9. No: a rectangle's sides need not all be equal.
10. 100 cubic metres

Section 3: Data, graphs and chance

1. 2 (4 x 1/4 = 1 and 2 x 1/2 = 1)
2. 6 (3 + 5 + 6 + 10 = 24, and 24 / 4 = 6)
3. 2 litres (1 + 1/4 + 3/4)
4. 1/2 litre (2 / 4 = 1/2)
5. Less: 2/5 = 4/10 and 1/2 = 5/10, and 4/10 is smaller.
6. 10
7. 41 (6 + 20 + 15)
8. 4 eggs (5 nests have 4 eggs, more than any other number)
9. 1 litre (3 x 1/2 = 6/4, and 2 x 1/4 = 2/4, so 6/4 - 2/4 = 4/4)
10. 2 litres (5 x 2/5 = 10/5 = 2), the same as the total

Section 4: Numbers and calculating

1. 34 and 0.34
2. Four and seventy-five thousandths
3. 0.62 > 0.6
4. 7.85
5. 9,342

6. 52
7. 16.25
8. 12.65
9. 7/12 (9/12 - 2/12)
10. 3/4 of a pizza
11. 1/12 (1/4 divided by 3 is 1/12)
12. Less than 5, because 2/3 is less than 1, so 5 groups of 2/3 make less than 5.